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A complete digital radio-frequency source for nuclear magnetic resonance spectroscopy

Jiang Yun, Jiang Yu, Tao Hongyan, and Li Gengying^{a)}

Department of Physics, Key Laboratory for Optical and Magnetic Resonance Spectroscopy, East China Normal University, Shanghai 200062, People's Republic of China

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A complete digital radio-frequency source suitable for broadband nuclear magnetic resonance applications is described. The use of a complete direct digital synthesis chip makes the design not only as a frequency- and phase-continuous tunable radio-frequency source, but also as an arbitrary wave form generator. This rf source is capable of generating a signal with frequency from dc up to 135 MHz directly. The whole design is mounted on a printed circuit board that can be inserted into one of the industry standard architecture slots in the personal computer and is controlled by the software. Details about the implementation and characteristics of the whole design are discussed in this article. © 2002 American Institute of Physics. [DOI: 10.1063/1.1497500]

I. INTRODUCTION

The radio-frequency (rf) source is an essential part in nuclear magnetic resonance (NMR) spectroscopy and magnetic resonance imaging (MRI). Modern NMR techniques such as selective excitation,¹ composite pulses,² and digital quadrature detection³ require that the rf source have the ability to rapidly switch its phase, frequency, and amplitude.

As the large-scale integrated circuit technology has progressed, more designs have been taken over by digital devices. The direct digital synthesis (DDS) technique is now becoming increasingly popular in building the frequency source for NMR. It uses digital data processing blocks as a means to generate a frequency- and phase-tunable output signal referenced to a fixed-frequency clock. Compared with the direct analog synthesis and the phase-locked loops, the DDS has the characteristic of faster frequency- and phase-continuous switching. In its simplest form, DDS consists of an address counter, a SINE look-up table ROM, and a digital to analog converter (DAC). Today the integration of high-speed DACs, DDS cores, amplitude modulation blocks, and the control interface into a single chip results in what is known as a complete DDS.

The frequency synthesizers from Programmed Test Sources, Inc. (Littleton, MA) which are popularly used in NMR spectrometers, are using DDS in their low-frequency sections. Several excellent designs and applications based on the DDS have been described, including those added to a normal frequency source to work as a phase shifter⁴⁻⁶ as well as those employing a DDS chip to work as a rf source in the spectrometer.^{3,7}

Herein a rf source with multifunctions suitable for the NMR spectrometer will be introduced. The present design employs a DDS chip (AD9854, Analog Device, Inc.), leading to the direct generation of rf output with frequency from dc up to 135 MHz. Moreover, it has the ability to rapidly

manipulate the instantaneous phase, frequency, and amplitude of the rf pulse. Based on the industry standard architecture (ISA) bus, which is available in most personal computers and implemented using a programmable logic device (PLD), makes this design much easier to be controlled. The whole design works not only as a frequency- and phase-tunable radio-frequency source, but as an arbitrary wave form generator as well.

II. HARDWARE ARCHITECTURE

In order to satisfy the requirements for most NMR experiments, we utilize two AD9854 chips on the printed circuit board. One is for ¹H and ¹⁹F NMR and the other is for other nuclei. A PLD works as a bridge between the AD9854 and the ISA bus. The block diagram of this rf board is shown in Fig. 1.

The AD9854 digital synthesizer is a highly integrated DDS device. The DDS core in the AD9854 provides 48-bit frequency resolution (1 μ Hz tuning steps) and 14-bit phase resolution (0.02° tuning steps). The maximum output frequency is 135 MHz ($0.45 \times f_{\text{CLOCK}}$, approximately) when referenced to the 300 MHz system clock.

The AD9854 is coupled with two internal high-speed quadrature DAC's. *I* (Cosine) and *Q* (Sine) DACs provide two complementary output, respectively, which enable four quadrature rf outputs from a single-AD9854 at the same time. For example, when the phase of one output of the *Q* DAC is 0°, the other output's phase of this DAC is 180°, and the phases of the *I* DAC outputs are 90° and 270°. In our design, one of these quadrature outputs is selected as a final output by a high speed 4:1 switching multiplexer (10 ns switch time, AD8184, Analog Device, Inc.).

The integration of the amplitude modulation (AM) block in AD9854 makes it possible to control the amplitude of rf outputs. A 12-bit register and a digital multiplier are integrated in the AM block. The value of this 12-bit register that is used to control the rf amplitude will be provided by the software running in the computer.

^{a)}Author to whom correspondence should be addressed; electronic mail: ligengying@netease.com

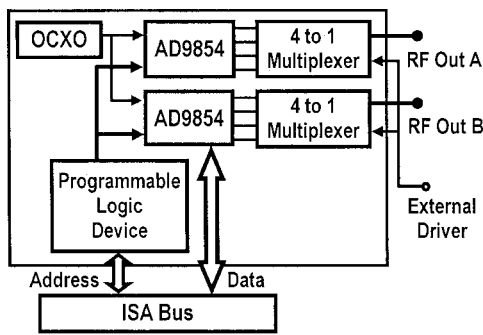


FIG. 1. A block diagram of the DDS rf source.

A complex programmable logic device (CPLD) (XC9572, Xilinx, Inc.) works as a bridge between the AD9854 and the ISA bus. The function of the PLD is to decode commands from the software and to control the status of the AD9854. Very high-speed integrated circuit hardware description language and schematic design flows are used in describing the function of the CPLD under the environment of Xilinx Foundation F1.5. The usage of CPLD not only simplifies the implementation of this rf source, but also makes the whole design flexible to any further improvements.

This rf source is immunized to parameter drift due to temperature. An on board 50 MHz oven-controlled crystal oscillator is supplied to both DDS as the frequency standard. Its stability is $5 \times 10^{-9}/\text{day}$. A programmable reference clock multiplier integrated in the AD9854 makes it easy to produce a 300 MHz internal system clock.

III. CONTROL SOFTWARE

A program developed using DELPHI 5 under WINDOW 98 OS is utilized to control the rf source card. The desired output frequency and phase set by the user will be translated to the frequency tuning word (FTW) and the phase tuning word (PTW) according to Eqs. (1) and (2) in the software

$$\text{FTW} = (\text{frequency} \times 2^N) \text{SYSCLK}, \quad (1)$$

where N is the frequency resolution (48 bits in present design), the frequency is expressed in Hertz, and SYSCLK is the system clock

$$\text{PTW} = (\text{phase} \times 2^M) / 2\pi, \quad (2)$$

where M is the phase resolution (14 bits in present design), FTW and PTW are decimal numbers. Once these two numbers have been calculated, they will be rounded to the integer and then converted to binary format in the software. These binary numbers will be written to DDS internal registers through the ISA bus.

The procedure of amplitude modulation is also controlled by software. In NMR, the amplitude modulation is written as

$$S_{\text{AM}} = A(t) \cdot \cos(2\pi f_c t + \theta_c), \quad (3)$$

where $A(t)$ is the modulating signal. $A(t)$ may be positive or negative depending on different types of functions. However, the 12-bit digital register in AD9854 is unable to express the negative value, and Eq. (3) becomes

$$S_{\text{AM}} = |A(t)| \cdot \cos(2\pi f_c t + \theta_c), \quad (4)$$

when the value of $A(t)$ is positive and

$$S_{\text{AM}} = |A(t)| \cdot \cos(2\pi f_c t + \theta_c + 180^\circ), \quad (5)$$

when the value of $A(t)$ is negative.

The modulating signal $A(t)$ created in the software can be any functions such as sin, cosine, sinc, Gauss, Lorentz, or the combination of these. According to the amplitude value of $A(t)$, the phase value of DDS rf output will be set correspondingly. In the software the absolute value of $A(t)$ and the phase value of DDS output will be stored in two arrays. These arrays' value will be transferred into the AD9854 one by one in every dwell time.

IV. EXPERIMENTS

Our rf source is broadband. Any frequency from dc to 135 MHz will be generated directly when the AD9854 is referenced to a 300 MHz clock. It is easy to obtain a much higher frequency by adding some rf stages (oscillator, mixer, and filter), or directly using the image output of DDS. The later one that has been reported on recently⁸ is much easier to use and more economic to get a higher frequency output.

In the present design, software is used to set the frequency, phase, and amplitude values of the DDS directly by the PC through the ISA bus. The CPLD decodes commands and gets the frequency, phase, and amplitude values from the software, and then sends these values to internal registers of AD9854. The delay for updating the frequency register value is about $14 \mu\text{s}$ with the CPU clock at 233 MHz. Updating the phase register and amplitude register requires $5 \mu\text{s}$ separately under the same condition. These delays are determined mainly by the transfer rate of the ISA bus.

An ultrafast switch between four quadrature rf phases is critical to the composite pulses or multiple pulses technique in the solid-state NMR experiment. Compared with other DDS chips mentioned before,⁸ the present AD9854 provides four quadrature rf outputs at the same time. With the 4:1 multiplexer, AD8184, a fast switch between four quadrature rf outputs can be achieved. The switching time between two outputs is 10 ns, which is determined by the channel switch-

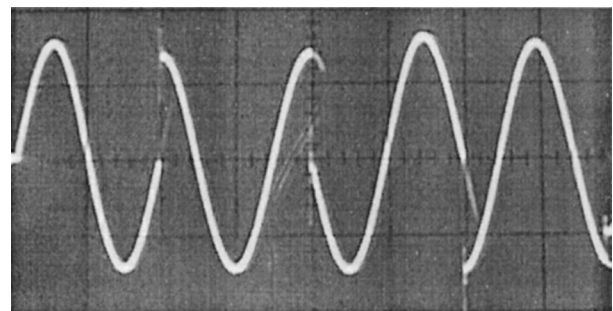


FIG. 2. Switching between the four most used phases 0° , 90° , 180° , and 270° . The frequency is 100 kHz with $5 \mu\text{s}/\text{div}$ in the oscilloscope.

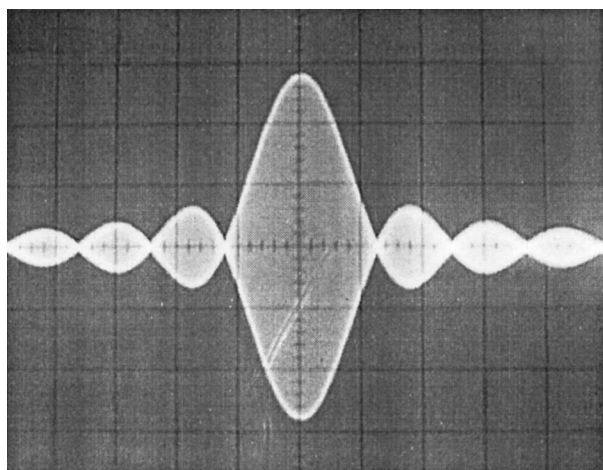


FIG. 3. Direct output of a $\text{sinc}(x)$ shaped rf pulse. The amplitude and phase is set every 10 us with 1 k data points.

ing time offered by AD8184. In this rf source board, AD8184 can be driven by an external TTL circuit. The switching time of the external driver determines the shortest duration of each channel. The switch between the four most used phases 0° , 90° , 180° , and 270° is shown in the oscilloscope traces of Fig. 2. For testing, the printer port in the personal computer is used as the external driver of the AD8184. The switching time of signals through the printer port is about 2 μs .

The internal amplitude modulation block in AD9854 enables this rf source to be an arbitrary wave form generator directly. An oscilloscope trace of DDS output modulated by $\text{Sinc}(x)$ is shown in Fig. 3.

The minimal dwell time is determined by the delay in updating the amplitude and phase of the DDS. In the present design, the total time of updating the amplitude and phase is within 10 μs directly by the PC through the ISA bus with the CPU clock at 233 MHz. This design is tested on a homebuilt spectrometer with a 2.1 T permanent magnet, derived from disassembling an out-of-date NMR spectrometer (FX-90Q, JOEL).

Figure 4 is the single pulse sequence ^1H -NMR spectrum obtained with this rf source board.

The experimental parameters are as follows: rf frequency $SF=90$ MHz, offset $O_1=11\,000$ Hz, 90° rf pulse width $D_1=5.0$ μs , dead time of the spectrometer $D_3=50$ μs , number of data points for free induction decay (FID) acquisition $TD=2048$ and $SW=2$ kHz, number of scans $NS=1$, pulse sequence repetition $D_0=5$ s. The FID was zero filling to 16 K before Fourier transform.

V. DISCUSSION

So far, we have described the complete digital rf source based ISA bus. Compared with the former design, our rf source has the following features. First, the present design

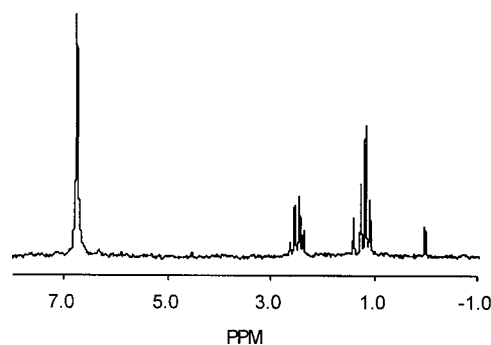


FIG. 4. ^1H -NMR spectrum of 1% ethyl benzene, obtained with the single pulse sequence.

has much broader bandwidth (from dc to 135 MHz). This is of great importance when one expects to do multinuclear observation in the NMR experiment. Second, the arbitrary wave form can be generated within the source. In a traditional transmitter, the rf wave form is achieved by adding an external analog multiplier or double-balance mixer. However, since the modulation is accomplished digitally in our design, it has many advantages such as better linearity and lower rf leakage. The resolution of the wave form is determined mainly by the digital multiplier (12 bits). Third, the ultrafast switch (about 10 ns) between four quadrature rf phases is provided and switching time between arbitrary phases is normally approximately 5 μs , in this ISA based strategy.

In summary, the use of the complete-DDS and the CPLD makes the present design not only multifunctional, but also easy to control and low cost. It is a complete excitation source for NMR, NQR, MRI, or related resonance technology. Although the slow transfer rate of the ISA bus limits the updating speed of the frequency-phase-amplitude value of DDS, the performance of the rf source may be improved by adding RAMs on board, using a faster transfer rate bus such as PCI, or directly using DSP to control the AD9854.⁹

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¹R. Freeman, Chem. Rev. **91**, 1397 (1991).

²M. H. Levitt and R. Freeman, J. Magn. Reson. **33**, 473 (1979).

³L. Gengying and X. Haibin, Rev. Sci. Instrum. **70**, 1511 (1999).

⁴J. Keeler and A. J. Shaka, J. Magn. Reson. **97**, 607 (1992).

⁵X. Wu, D. A. Patterson, and L. G. Butler, Rev. Sci. Instrum. **64**, 1235 (1993).

⁶E. Sternin, Rev. Sci. Instrum. **66**, 3144 (1995).

⁷F. Momo, A. Sotgiu, L. Testa, and A. Zanin, Rev. Sci. Instrum. **65**, 3291 (1994).

⁸L. Gengying, J. Yu, Y. Xiaolong, and J. Yun, Rev. Sci. Instrum. **72**, 4460 (2001).

⁹The schematic circuit diagram and the printed circuit board are available at: <http://www.anac.ecnu.edu.cn/dds>